THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 12	Take the lowest powers of common primes
2	**(c)** 29	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)$
3	**(d)** -2	Eliminate y
4	**(d)** $\frac{9}{2}$	Sum $=-b/a$
5	**(a)** 40	First find $d=3$
6	**(c)** 4	$AE=k\times AD$
7	**(b)** 24	Find the hypotenuse
8	**(b)** $\frac{3}{4}$	Substitute the standard values
9	**(c)** 20 m	Add the height of the eyes: $2+18$; Height of building $=20$ m
10	**(c)** 30 cm	$r^2=OP^2-PT^2=34^2-16^2=900$; $r=30$ cm
11	**(c)** 56 cm²	Circle $=154$ cm²; square $=98$ cm²; Difference $=56$ cm²
12	**(c)** 225.5 m²	Canvas = CSA of cylinder + CSA of cone; $=88+137.5=225.5$ m²
13	**(d)** 32.4	$\sum f_i=50$ and $\sum f_ix_i=1620$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{1620}{50}=32.4$
14	**(c)** $\frac{1}{4}$	Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $=9$; $P=\frac{9}{36}=\frac{1}{4}$
15	**(a)** 24	Take the highest powers of all primes
16	**(a)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
17	**(d)** 5	Eliminate x
18	**(d)** 2	Product $=c/a$
19	**(a)** 24	$n=\frac{l-a}{d}+1$
20	**(c)** 5	Perimeter ratio = side ratio

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Numbers exactly divisible: 240, 1024 (1); HCF $=16$ (1).	1+1
22	$\alpha+\beta=8$, $\alpha\beta=12$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=40$ (1).	1+1
23	Write as $7x+2y+6=0$, $3x+5y-14=0$ (1); $\frac{x}{-58}=\frac{y}{116}=\frac{1}{29}$ (1); $x=-2$, $y=4$ (1).	1+1+1
24	$D=(-9)^2-4(2)(4)=49$ (1); $x=\frac{9\pm7}{4}$ (1); roots $\frac{1}{2}, 4$ (1).	1+1+1
25	$a_n=a+(n-1)d$: $127=7+(n-1)(6)$ (1); $n-1=20$ (1); $n=21$, so it is the 21st term (1).	1+1+1
26	$\frac{AD}{AB}=\frac{AE}{AC}$ (1); $AE=\frac{3\times10}{5}=6$ cm (1); $EC=10-6=4$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	HCF × LCM = product of numbers (1); other $=\frac{9\times360}{72}$ (1) $=45$ (1).	1+1+1
28	$\alpha+\beta=6$, $\alpha\beta=8$ (1); $\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{4}$ (1); $\alpha^2+\beta^2=20$ (1).	1+1+1
29	Multiply (i) by 5 and (ii) by 2 and subtract: $(29)x=-58$ (2); $x=-2$ (1); put in (i): $y=4$ (1); check in (ii) (1).	2+1+1+1
30	Multiply by $x(x+1)$: $2x+1=\frac{5}{6}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x=2$ (1).	1+1+1
31	$a+2d=19$, $a+7d=54$ (2); subtracting, $d=7$, $a=5$ (2); $a_{15}=103$ (1).	2+2+1
	OR: $\frac{AD}{AB}=\frac{AE}{AC}$ (1); $AE=\frac{4\times18}{9}=8$ cm (1); $EC=18-8=10$ cm (1).	1+1+1

32	$AB^2 = 16$, $BC^2 = 13$, $CA^2 = 13$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
	OR: $\frac{1}{2} = \sin 30^\circ$ (1); $3x + 0 = 30$ (1); $x = 10^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Capacity $=$ HCF$(850, 680) = 170$ L (3); measures 5 and 4 (2).	3+2
34	Sum of the three zeroes $= -\frac{b}{a} = 6$ (2); third zero $= 6 - (1) - (2) = 3$ (2); zeroes $1, 2, 3$ (1).	2+2+1
35	Son x, father $10x$ (1); $10x + 6 = 4(x + 6)$ (2); $x = 3$ (1); son 3 years, father 30 years (1).	1+2+1+1
36	Speed x: $\frac{200}{x} - \frac{200}{x+5} = 2$ (1); $1000 = 2x(x + 5)$ so $x^2 + 5x - 500 = 0$ (2); $(x - 20)(x + 25) = 0$ (1); speed 20 km/h (1).	1+2+1+1
	OR: $S_n = \frac{n}{2}[2(54) + (n - 1)(-4)] = 360$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
37	Draw a horizontal line from the top of the shorter pole: right triangle with legs 9 and 12 (2); $x^2 = 81 + 144$ (1); $x = 15$ m (2).	2+1+2
	OR: Let ratio $m : n$; $y = \frac{m(4)+n(-5)}{m+n} = 0$ (2); $m : n = \frac{5}{4} : 1$, i.e. 5:4 (1); point $x = \frac{5(6)+4(-3)}{9} = 2$ (2).	2+1+2
38	LHS $= \left(\frac{1 - \sin A}{\cos A}\right)^2$ (2); $= \frac{(1 - \sin A)^2}{1 - \sin^2 A}$ (1); $= \frac{1 - \sin A}{1 + \sin A}$ (2).	5
	OR: Let height $= h$; distances $h \cot 60^\circ$ and $h \cot 30^\circ$ add up to 60 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 60$ gives $h = 15\sqrt{3}$ (2); Height $15\sqrt{3}$ m; distances 15 m and 45 m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	Midpoint
2	**(c)** 3	$\frac{c}{q}+\frac{p}{q}$
3	**(d)** $4\sqrt{3}$ m	$h(\cot 30^\circ-\cot 60^\circ)=8$ gives $h=8\div(\frac{2\sqrt{3}}{3})$; Height $=4\sqrt{3}$ m (distances 12 m and 4 m)
4	**(a)** 92 cm	Perimeter $=PA+AO+OB+BP=2(16)+2(30)$; Perimeter $=92$ cm
5	**(b)** 672 cm²	Circle area $=\frac{22}{7}\times 28^2=2464$ cm²; square $=3136$ cm²; Left over $=672$ cm²
6	**(c)** 396 cm³	$\frac{2}{3}\pi(R^3-r^3)=\frac{2}{3}\times\frac{22}{7}(6^3-3^3)$; Volume $=396$ cm³
7	**(a)** 26	Median class 20–30: $l=20$, $cf=16$, $f=15$, $h=10$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=20+\frac{25-16}{15}\times 10=26$
8	**(a)** $\frac{7}{8}$	Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $=7$; $P=\frac{7}{8}$
9	**(c)** 13	HCF of 65 and 117
10	**(b)** 10	$2^2-7(2)+k=0$
11	**(c)** 2	All ratios equal
12	**(b)** 7	Roots are 2, 9
13	**(a)** 1665	AP of two-digit multiples
14	**(c)** 6	$h^2=pq$
15	**(b)** 4	$\frac{x_1+x_2+x_3}{3}$
16	**(b)** $\frac{1}{2}$	$\frac{c}{p}-\frac{q}{p}$
17	**(d)** 24 m	The points are on opposite sides, so the distances add; Distance $=24$ m
18	**(d)** 360 cm²	Area $=2\times\frac{1}{2}\times 40\times 9$; Area $=360$ cm²
19	**(c)** 1034 m²	Area $=\pi(R^2-r^2)=\frac{22}{7}(27^2-20^2)$; Area of path $=1034$ m²
20	**(a)** 6835.5 cm³	Cube $=9261$ cm³; hemisphere of radius 10.5 cm $=\frac{2}{3}\pi r^3=2425.5$ cm³; Volume $=6835.5$ cm³

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$(6)^2+(y-2)^2=100$ (1); $(y-2)^2=64$ (1); $y=2\pm 8$, i.e. $y=10$ or -6 (1).	1+1+1
22	Divide by $\cos A$: $\frac{\tan A+1}{\tan A-1}$ (1) $=7$ (1).	1+1
23	Horizontal distance $d=6\cot 60^\circ=2\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $=d\tan 30^\circ=2$ m (1); Tower $=8$ m (1).	1+1
24	$PA=PB$, so $\triangle PAB$ is isosceles and $\angle PBA=\angle PAB$; $\angle APB=180^\circ-30^\circ-30^\circ$ (1); $\angle APB=120^\circ$ (1).	1+1
25	In 15 minutes the hand turns through $6^\circ\times 15=90^\circ$; Area $=\frac{90}{360}\times\frac{22}{7}\times 28^2$ (1); Area $=616$ cm² (1).	1+1
26	Volume of sphere $=$ volume of cone; $\frac{4}{3}\pi R^3=\frac{1}{3}\pi(6)^2(3)$ gives $R^3=27$ (1); $R=3$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let height $=h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 4 m (1); $h(\cot 45^\circ-\cot 60^\circ)=4$ gives $h=4\div(1-\frac{\sqrt{3}}{3})$ (1); Height $=6+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m) (1).	1+1+1
28	Tangents from an external point are equal: $QA=QB$ (1); $9x+4=5x+20$ gives $x=4$ (1); $QA=9(4)+4=40$ cm (1).	1+1+1
29	Circumference $=2\pi r=2\times\frac{22}{7}\times 14=88$ m (1); Cost $=88\times 12$ (1); Cost $=$ ₹ 1056 (1).	1+1+1

30	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 4158$ cm^3; two hemispheres $= 4851$ cm^3 (1); Volume $= 9009$ cm^3 (1).	1+1+1
31	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 50$ and $\sum f_i u_i = -48$ (1); Mean $= a + h\,\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-48}{50} = 25.4$ (1).	1+1+1
	OR: $P(\text{blue}) = \frac{x}{7+x}$ and $P(\text{red}) = \frac{7}{7+x}$ (1); $\frac{x}{7+x} = 2 \times \frac{7}{7+x}$ gives $x = 2 \times 7$ (1); $x = 14$ (1).	1+1+1
32	Required time = LCM$(12, 15, 18)$ (1); $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$ (1); LCM $= 180$ minutes (1).	1+1+1
	OR: Sum of new zeroes $= \dfrac{\alpha + \beta}{\alpha\beta} = \frac{5}{6}$ (1); product $= \dfrac{1}{\alpha\beta} = \frac{1}{6}$ (1); polynomial $x^2 - (\frac{5}{6})x + (\frac{1}{6})$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{52^2 - 20^2} = 48$ cm; perimeter $= 2(48 + 20)$ (2); Perimeter 136 cm; area $= 2 \times \frac{1}{2} \times 20 \times 48 = 960$ cm^2 (1).	2+2+1
34	Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{45}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 57.75$ cm^2 (1).	2+2+1
35	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(20) = \pi(4)^2 x$ (2); $x = 15$ cm (1).	2+2+1
36	$\sum f_i = 45 + p$ and $\sum f_i x_i = 1445 + 35p$ (2); $\frac{1445+35p}{45+p} = 32.4$ gives $(35 - 32.4)p = 13$ (2); $p = 5$ (1).	2+2+1
	OR: Total outcomes $= 36$ (2); Favourable outcomes: 4; 15; 12 (2); Probabilities: $\frac{1}{9}$, $\frac{5}{12}$, $\frac{1}{3}$ (1).	2+2+1
37	Side = HCF$(24, 16) = 8$ m (2); tiles $= \dfrac{24 \times 16}{64} = 6$ (3).	2+3
	OR: Sum of zeroes $= -\dfrac{b}{a} = \dfrac{12}{k} = 3$ (2); $k = 4$ (1); product $= \dfrac{c}{a} = \dfrac{8}{4} = 2$ (2).	2+1+2
38	Let fraction $\frac{x}{y}$: $2(x + 1) = 1(y + 1)$ and $9(x + 2) = 5(y + 2)$ (2); simplify and solve (2); $x = 3$, $y = 7$, fraction $\frac{3}{7}$ (1).	2+2+1
	OR: Base x, altitude $x - 1$: $x^2 + (x - 1)^2 = 841$ (1); $2x^2 - 2x - 840 = 0$, i.e. $x^2 - x - 420 = 0$ (2); $(x - 21)(x + 20) = 0$ (1); base 21 cm, altitude 20 cm (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 77	$l = 70,\ f_1 = 16,\ f_0 = 2,\ f_2 = 10,\ h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$
2	**(b)** $\frac{7}{15}$	Favourable numbers: 3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30; number $= 14$; $P = \frac{14}{30} = \frac{7}{15}$
3	**(b)** 27	LCM + remainder
4	**(d)** 3	$2^2 + 2k - 10 = 0$
5	**(c)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
6	**(a)** 20	$16 - 36 + k = 0$
7	**(d)** 623	Use S_n
8	**(c)** 36	Similar triangles
9	**(a)** 5	Distance from centre
10	**(a)** 2	Each ratio is 1
11	**(b)** $3\sqrt{3}$ m	Distance from the foot $= 3\cot 30^\circ = 3\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 3\sqrt{3}$ m
12	**(a)** 96 cm	Half-chord $= \sqrt{52^2 - 20^2} = 48$ cm; Chord $= 2 \times 48 = 96$ cm
13	**(d)** 77 cm²	$\frac{60}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 77$ cm²
14	**(c)** 256 cm	$\frac{4}{3}\pi(12)^3 = \pi(3)^2 L$; $L = 256$ cm
15	**(d)** 6	$\frac{1750 + 35p}{44 + p} = 39.2$ gives $(35 - 39.2)p = -25.2$; $p = 6$
16	**(c)** 6	$\frac{x}{3+x} = 2 \times \frac{3}{3+x}$ gives $x = 2 \times 3$; $x = 6$
17	**(a)** non-terminating repeating	$210 = 2 \times 3 \times 5 \times 7$
18	**(a)** -7	$p = -(\alpha + \beta) = -(1 + 6)$
19	**(d)** 9	Add the two equations
20	**(a)** two distinct real roots	$D = 4$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -13$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-13}{50} = 32.4$ (1).	1+1
22	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,2), (1,5), (2,1), (2,4), (3,3), (3,6), (4,2), (4,5), (5,1), (5,4), (6,3), (6,6); number $= 12$ (1); $P = \frac{12}{36} = \frac{1}{3}$ (1).	1+1
23	LCM$(12, 16, 18) = 144$ (1); number $= 144 + 5 = 149$ (1).	1+1
24	$p(3) = 0$: $-15 + k = 0$ (1) so $k = 15$ (1).	1+1
25	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{6} \neq \frac{2}{1}$, i.e. $k \neq 12$ (1).	1+1
26	Put $x = 4$: $16 - 36 + k = 0$ (1); $k = 20$ (1); other root $= 9 - 4 = 5$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let tens digit x, ones digit y: $x + y = 11$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 3$, $y = 8$, number 38 (1).	1+1+1
28	$D = 25$ (1); two distinct real roots (1); roots $2, \frac{1}{3}$ (1).	1+1+1
29	$a_1 = 5$, $a_{20} = 62$ (1); $d = 3$ (1); $S_{20} = \frac{20}{2}(5 + 62) = 670$ (1).	1+1+1
30	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 2$ (1); $QR = 8$ cm (1); $RP = 10$ cm (1).	1+1+1
31	Section formula $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$ (1); $x = \frac{2(8) + 3(-2)}{5} = 2$ (1); $y = 7$ (1).	1+1+1

	OR: Hypotenuse $= \sqrt{15^2 + 8^2} = 17$ (1); $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$ (1); $\sec\theta = \frac{17}{8}$ (1).	1+1+1
32	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 6\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 6\sqrt{3}$, so $d = 9$ (1); $H = d\tan 60^\circ = 9\sqrt{3}$; tower $9\sqrt{3}$ m, distance 9 m (1).	1+1+1
	OR: For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $18 + 12 = AD + 15$ (1); $AD = 15$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	AP with $a = 45$, $l = 25$, $n = 11$ (1); $S_n = \frac{11}{2}(45 + 25) = 385$ cm (2); middle term $= \frac{45+25}{2} = 35$ cm (2).	1+2+2
34	Diagonals bisect at right angles: half-diagonals 6 and 8 (1); $s^2 = 6^2 + 8^2 = 100$ (1); side $= \sqrt{100}$ cm (1).	1+1+1
35	Let the point be $(x, 0)$; $(x-1)^2 + 25 = (x-7)^2 + 9$ (2); $x = \frac{8}{3}$ (2); point $(\frac{8}{3}, 0)$ (1).	2+2+1
36	Divide numerator and denominator by $\cos A$ (1): $\frac{3\tan A + 2}{4\tan A + 1}$ (1); $= \frac{3(\frac{15}{8}) + 2}{4(\frac{15}{8}) + 1} = \frac{61}{68}$ (2); verified with $\sin A = \frac{15}{17}$, $\cos A = \frac{8}{17}$ (1).	1+1+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 2 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 2$ gives $h = 2 \div (\sqrt{3} - 1)$ (2); Height $= 1 + \sqrt{3}$ m (distances $3 + \sqrt{3}$ m and $1 + \sqrt{3}$ m) (1).	2+2+1
37	$OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$ (2); In quadrilateral $OAPB$ the angle sum is 360° (1); $\angle APB + \angle AOB = 360^\circ - 90^\circ - 90^\circ = 180^\circ$ (2).	2+1+2
	OR: The 8 ribs divide the circle into 8 equal sectors (2); Area of the circle $= \frac{22}{7} \times 42^2 = 5544$ cm^2 (2); Area between ribs $= 5544 \div 8 = 693$ cm^2 (1).	2+2+1
38	Volume of water is the same in both tanks (2); $\pi r^2 h = 3080$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 3080 \div 308 = 10$ cm (1).	2+2+1
	OR: Median class is 50–60 with $f = 20$, and the cumulative frequency before it is $cf = 21 + x$ (2); $54 = 50 + \frac{50-(21+x)}{20} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 63 - x = 16$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.